

QA Overview (TG 40/142/MPPG 8a/198).

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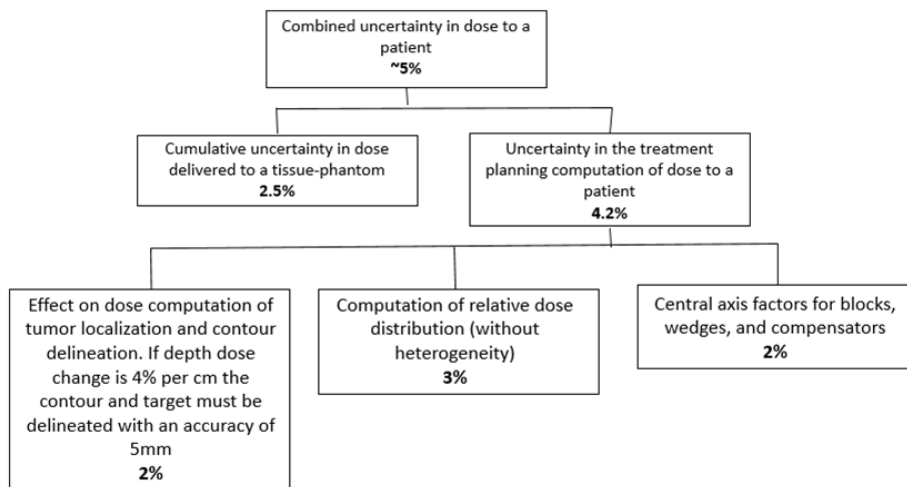


Content Table

TG-40	TG-142	MPPG 8a	TG-198
1994	2009	2017	2021
Comprehensive QA for radiation oncology	QA of Medical Accelerators	Linear Accelerator Performance Tests	Implementation Guide for TG-142
QA for ^{60}Co , Medical Accelerators, Simulators, Measurement Devices, TPS, individual patient QA, Brachytherapy, etc.	(non-IMRT, IMRT, SRS/SBRT), MLCs, Asymmetric Jaws, Image Guidance, Respiratory Gating, TBI	(non-IMRT, IMRT, SRS/SBRT), MLCs, Asymmetric Jaws, Image Guidance, Respiratory Gating, TBI	Objectives, tolerances and methods & equipment for: (non-IMRT, IMRT, SRS/SBRT), MLCs, Asymmetric Jaws, Image Guidance, Respiratory Gating, TBI, TSE, VMAT
Table II Safety / Mechanical / Dosimetry	Tables I to VI Safety / Mechanical / Dosimetry / Gating / EDW / MLC / Imaging	Tables 2 to 6 Safety / Mechanical / Dosimetry / Wedge / Annual Review	Tables I, III, IV-VII Daily / Monthly / Annual / Wedge / MLC / Imaging

General Recommendations

- ICRU recommendation of delivered dose to be within $\pm 5\%$, 5mm of the prescribed dose (Not a regulation, but a guideline for QMPs to interpret).



- Specific test recommendations not the goal of any of the three reports. Describes the qualities of a "good" test.
 - E.g. Measurements precise enough that $2\sigma < \text{tolerance}$

Why TG-142?

- New Treatment Technologies:
 - MLCs
 - Asymmetric Jaws
 - Dynamic/Virtual Wedges
 - Electronic portal imaging devices (EPIDs)
 - Image Guidance (CBCT/kV)
 - Respiratory Gating
- New Role of Treatment Units:
 - Accuracy and Precision for non-IMRT, IMRT, and SRS/SBRT should be considered individually

What does TG-198 do -> Implements TG-142?

- Serve as a reference on how to perform the TG-142 recommended tests
- Provide estimate of the range of time, appropriate personnel, and qualifications necessary to complete the tests in TG-142
- Provide sample daily, weekly, monthly or annual quality assurance forms
- Adds several tests and criteria specific to VMAT not specifically addressed by TG-142

Action Levels:

Level 1: Inspection Action

- Sudden and significant deviations (even if tolerance is not exceeded)
- Example: A change in personnel, setup, or maintenance causes a measurement shift

Level 2: Scheduled Action

- Example: Consecutive QA results at or near tolerance or a single result that exceeds tolerance but not excessively

Level 3: Immediate Action/Stop Treatment Action

- Example: Non-functional safety interlock or excessive error in dosimetry parameters

★ Daily QA:

- Tests that can seriously affect treatment accuracy
 - Patient position
 - Patient dose
 - Safety
- Include parameters that can affect dose delivery by
 - Dosimetric (output constancy)
 - Geometric (laser, ODI, field size)
- Performed typically by morning warmup therapist
 - Trained by QMP with well-defined policy and procedure

TG-40



TABLE I. Daily.				
Procedure	Machine-type tolerance			
	Non-IMRT	IMRT	SRS/SBRT	
Dosimetry				
X-ray output constancy (all energies)				
Electron output constancy (weekly, except for machines with unique e-monitoring requiring daily)		3%		Same
Mechanical				
Laser localization	2 mm	1.5 mm	1 mm	2 mm
Distance indicator (ODI) @ iso	2 mm	2 mm	2 mm	Same
Collimator size indicator	2 mm	2 mm	1 mm	2 mm
Safety				
Door interlock (beam off)		Functional		Same
Door closing safety		Functional		None
Audiovisual monitor(s)		Functional		Same
Stereotactic interlocks (lockout)	NA	NA	Functional	Same
Radiation area monitor (if used)		Functional		
Beam on indicator		Functional		None

- Daily Imaging**
 - Trained by QMP with well-defined policy and procedure

Daily^a

Planar kV and MV (EPID) imaging

Collision interlocks	Functional	Functional
Positioning/repositioning	≤2 mm	≤1 mm
Imaging and treatment coordinate coincidence (single gantry angle)	≤2 mm	≤1 mm

Cone-beam CT (kV and MV)

Collision interlocks	Functional	Functional
Imaging and treatment coordinate coincidence	≤2 mm	≤1 mm
Positioning/repositioning	≤1 mm	≤1 mm

★ Daily Equipment

- Daily QA 3 by Sun Nuclear
 - 13 ion chambers
 - 5 ion chambers for flatness/symmetry
 - 4 ion chambers for electron energy
 - 4 ion chambers for photon energy
 - 12 diodes
 - Light-radiation coincidence
- Isocube
 - Central BB, offset fiducials
 - Laser localization
 - Positioning/repositioning

★ Monthly QA:

- Verify parameters that have a
 - Lower likelihood of changing on a day to day basis

- Smaller impact on patient treatment quality
- Tolerances tighter than for daily QA, because
 - Dosimetry devices calibrated by Accredited Dosimetry Calibration Laboratory (ADCL)
 - Performed by QMP
- Best practice order of operations (gantry head → phantom)
 - Safety
 - Mechanical
 - Imaging
 - Dosimetry

• Monthly Mechanical

• Monthly Dosimetry

TABLE II. Monthly.				TG-40	
Procedure	Machine-type tolerance				
	Non-IMRT	IMRT	SRS/SBRT		
Dosimetry					
X-ray output constancy				Same	
Electron output constancy		2%			
Backup monitor chamber constancy					
Typical dose rate ^a output constancy	NA	2% (@ IMRT dose rate)	2% (@ stereo dose rate, MU)		
Photon beam profile constancy		1%		2%	
Electron beam profile constancy		1%		3%	
Electron beam energy constancy		2%/2 mm		2mm	2 %

• Monthly Safety/Gating

TABLE II. Monthly.				TG-40	
Procedure	Machine-type tolerance				
	Non-IMRT	IMRT	SRS/SBRT		
Safety					
Laser guard-interlock test		Functional			
Respiratory gating					
Beam output constancy	2%				
Phase, amplitude beam control	Functional				
In-room respiratory monitoring system	Functional				
Gating interlock	Functional			None	
				N/A	

• Monthly Imaging

Procedure	Application-type tolerance		
	non-SRS/SBRT	SRS/SBRT	
Monthly			
Planar MV imaging (EPID)			
Imaging and treatment coordinate coincidence (four cardinal angles)	≤2 mm	≤1 mm	2 mm
Scaling ^b	≤2 mm	≤2 mm	2.5 mm
Spatial resolution	Baseline ^c	Baseline	Same
Contrast	Baseline	Baseline	Same
Uniformity and noise	Baseline	Baseline	Same
Planar kV imaging ^d			
Imaging and treatment coordinate coincidence (four cardinal angles)	≤2 mm	≤1 mm	2 mm
Scaling	≤2 mm	≤1 mm	Same Same Same Same
Spatial resolution	Baseline	Baseline	
Contrast	Baseline	Baseline	
Uniformity and noise	Baseline	Baseline	
Cone-beam CT (kV and MV)			
Geometric distortion	≤2 mm	≤1 mm	2 mm
Spatial resolution	Baseline	Baseline	Same
Contrast	Baseline	Baseline	Same
HU constancy	Baseline	Baseline	Same
Uniformity and noise	Baseline	Baseline	Same



Annual QA:

Subset of the tests performed during acceptance and commissioning
Baselines established, reconfirmed, or updated
Performed by QMP

PCAM standards used:

- Published department data
- AAPM TG-40/142
- Original Acceptance Test Data
- Varian Acceptance Test Specifications
- PA State Regulations

Annual Mechanical/Safety:

TABLE III. Annual.				TG-40	
Machine-type tolerance					
Procedure	Non-IMRT	IMRT	SRS/SBRT		
Mechanical					
Collimator rotation isocenter		±1 mm from baseline			Same
Gantry rotation isocenter		±1 mm from baseline			Same
Couch rotation isocenter		±1 mm from baseline		2 mm diameter	Same
Electron applicator interlocks		Functional			Same
Coincidence of radiation and mechanical isocenter	±2 mm from baseline	±2 mm from baseline	±1 mm from baseline		2 mm
Table top sag		2 mm from baseline			
Table angle		1°		None	Same
Table travel maximum range movement in all directions		±2 mm		Only vertical	Same
Stereotactic accessories, lockouts, etc.	NA	NA	Functional	None	Same
Safety					
Follow manufacturer's test procedures		Functional			Same
Respiratory gating					
Beam energy constancy		2%			
Temporal accuracy of phase/amplitude gate on		100 ms of expected		None	N/A
Calibration of surrogate for respiratory phase/amplitude		100 ms of expected			
Interlock testing		Functional			

Annual Dosimetry:

TABLE III. Annual.				TG-40	
Machine-type tolerance					
Procedure	Non-IMRT	IMRT	SRS/SBRT		
Dosimetry					
X-ray flatness change from baseline		1%		N/A	1%
X-ray symmetry change from baseline		±1%			2% Abs
Electron flatness change from baseline		1%			2%
Electron symmetry change from baseline		±1%			3% Abs
SRS arc rotation mode (range: 0.5-10 MU/deg)	NA	NA	Monitor units set vs delivered: 1.0 MU or 2% (whichever is greater) Gantry arc set vs delivered: 1.0° or 2% (whichever is greater)		
X-ray/electron output calibration (TG-51)		±1% (absolute)		2%	2%
Spot check of field size dependent output factors for x-ray (two or more FSs)		2% for field size <4×4 cm ² , 1% ≥4×4 cm ²		2%	1%
Output factors for electron applicators (spot check of one applicator/energy)		±2% from baseline			2%
X-ray beam quality (PDD ₁₀₀ or TMR ₁₀₀)		±1% from baseline			PDD: 1%, other 2%
Electron beam quality (R ₉₀)		±1 mm			R50 1mm, other 2mm
Physical wedge transmission factor constancy		±2%			None
X-ray monitor unit linearity (output constancy)	±2% ≥5 MU	±5% (2-4 MU), ±2% ≥5 MU	±5% (2-4 MU), ±2% ≥5 MU	1%	1%
Electron monitor unit linearity (output constancy)		±2% ≥5 MU			
X-ray output constancy vs dose rate		±2% from baseline		N/A	2%
X-ray output constancy vs gantry angle		±1% from baseline		2%	1%
Electron output constancy vs gantry angle		±1% from baseline		2%	1%
Electron and x-ray off-axis factor constancy vs gantry angle		±1% from baseline		2%	1%
Arc mode (expected MU, degrees)		±1% from baseline			
TBI/TSET mode		Functional		None	Same
PDD or TMR and OAF constancy		1% (TBI) or 1 mm PDD shift (TSET) from baseline			Same
TBI/TSET output calibration		2% from baseline			Same
TBI/TSET accessories		2% from baseline			Same

Annual Imaging:

TABLE VI. Imaging.			
Application-type tolerance			
Procedure	non-SRS/SBRT	SRS/SBRT	Radiation Safety ↓
Annual (A)			
Planar MV imaging (EPID)			
Full range of travel SDD	±5 mm	±5 mm	
Imaging dose ^a	Baseline	Baseline	
Planar kV imaging			
Beam quality/energy	Baseline	Baseline	
Imaging dose	Baseline	Baseline	
Cone-beam CT (kV and MV)			
Imaging dose	Baseline	Baseline	

*Annual review of imaging dose deemed sufficient due to minimal effect on overall dose and existing daily/monthly reviews of parameters that could affect dose

**††Dana Radiation Safety responsible for tube and x-ray source calibration checks

Annual VMAT guidelines (TG-218):

- Ensures linac can deliver VMAT plans that pass institutional VMAT QA criteria based on TG218 guidelines.
- Tests accuracy of VMAT delivery when treatment is interrupted and restarted

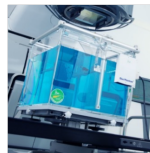
VMAT				
Test patient-specific VMAT QA constancy	Plan passes QA at a pass rate $\geq$ baseline pass rate minus 2% AND Plan passes institutional VMAT QA criteria	VMAT QA phantom	20–30 min	QMP
Interruption test	$\pm$ 1% deviation from uninterrupted passing rate	VMAT QA phantom	10–15 min	QMP

Annual Equipment:

► Blue Phantom 2 by IBA Dosimetry

- Large 3D water tank with robotic arms

Exterior water tank	
dimensions (LxWxH):	675 mm x 645 mm x 560 mm
Scanning volume (LxWxH):	480 mm x 480 mm x 410 mm
Position resolution:	0.1 mm
Position accuracy:	$\pm$ 0.1 mm
Position reproducibility:	$\pm$ 0.1 mm
Scanning speed:	max 50 mm/s
Approximate volume:	200 l
Wall thickness / material:	15 mm / acrylic
Weight:	40 kg



► 1D Scanner Water Tank by Sun Nuclear

- Smaller 1D scanner for output factors/TG-51



► Radiochromic Film

- Star Patterns to test isocentricity

Annual MLC/QA:

TABLE V. Multileaf collimation (with differentiation of IMRT vs non-IMRT machines).

Procedure	Tolerance	
	Weekly (IMRT machines)	
Qualitative test (i.e., matched segments, aka "picket fence")	Visual inspection for discernable deviations such as an increase in interleaf transmission	
	Monthly	
Setting vs radiation field for two patterns (non-IMRT)	2 mm	
Backup diaphragm settings (Elekta only)	2 mm	
Travel speed (IMRT)	Loss of leaf speed >0.5 cm/s	
Leaf position accuracy (IMRT)	1 mm for leaf positions of an IMRT field for four cardinal gantry angles. (Picket fence test may be used, test depends on clinical planning-segment size)	
	Annually	
MLC transmission (average of leaf and interleaf transmission), all energies	$\pm 0.5\%$ from baseline	
Leaf position repeatability	± 1.0 mm	
MLC spoke shot	≤ 1.0 mm radius	
Coincidence of light field and x-ray field (all energies)	± 2.0 mm	
Segmental IMRT (step and shoot) test	<0.35 cm max. error RMS, 95% of error counts <0.35 cm	
Moving window IMRT (four cardinal gantry angles)	<0.35 cm max. error RMS, 95% of error counts <0.35 cm	



[Picket fence](#)
[G0, 90, 180 270](#)

[DLG 0.6mm,](#)
[trans 0.5%](#)

[Same](#)

[Same](#)

[Same](#)

Wedge QA:

TABLE IV. Dynamic/universal/virtual wedges.

Dynamic-including EDW (Varian), virtual (Siemens), universal (Elekta) wedge quality assurance				
		Tolerance		
Frequency	Procedure	Dynamic	Universal	Virtual
Daily	Morning check-out run for one angle		Functional	
Monthly	Wedge factor for all energies	C.A. axis 45° or 60° WF (within 2%) ^a	C.A. axis 45° or 60° WF (within 2%) ^a	5% from unity, otherwise 2%
Annual	Check of wedge angle for 60°, full field and spot check for intermediate angle, field size	Check of off-center ratios @ 80% field width @ 10 cm to be within 2%		



[60° \(2%\)](#)

[Off axis ratio](#)
[2%](#)

^aRecommendation to check 45° if angles other than 60° are used.

TG-40 Dosimetry Equipment Procedures:

QA of Dosimetry Equipment	Frequency	Tolerance
Ion Chambers		
ADCL Calibration (if for reference dosimetry)	2 years	Documented
Redundancy check	Every use	2%
Linearity	2 years	0.5%
Extra-cameral signal (stem effect)	Initially or after repair	0.5%
Leakage	Every use	0.1%
Recombination	Initially or after repair	
Film		

Calibration	Each batch	Documented
Processor uniformity	Every use	Documented
TLD		
Calibration	Every use	Documented
Linearity	Each batch	Documented
Diodes		
Calibration	Monthly	2%
Energy dependence	Initially	Documented
Extra-cameral signal (stem effect)	Initially	Documented
Linearity	Initially	Documented
Accessories		
Thermometer	Initially	0.1 °C
Barometer	Every 3 months	1 mmHg